UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

| | |
|---|---|
| COURSE | COMPUTER ORGANIZATION AND ARCHITECTURE |
| COURSE CODE | CSC429 |
| EXAMINATION | NOVEMBER 2026 |
| TIME | 3 HOURS |

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

«:> Hak Cipta Universiti Teknologi MARA 

**PART A (80 MARKS)**

Answer ALL questions in this part.

**QUESTION 1** (5 marks)

Given the following Non Uniform Memory Access (NUMA) diagram.

a) State ONE (1) advantage of Symmetric Mufti-Processor (SMP) that is applied in NUMA architecture.

b) Describe ONE (1) disadvantage of cluster architecture.

c) Draw a possible architecture for cluster architecture based on Figure 1.

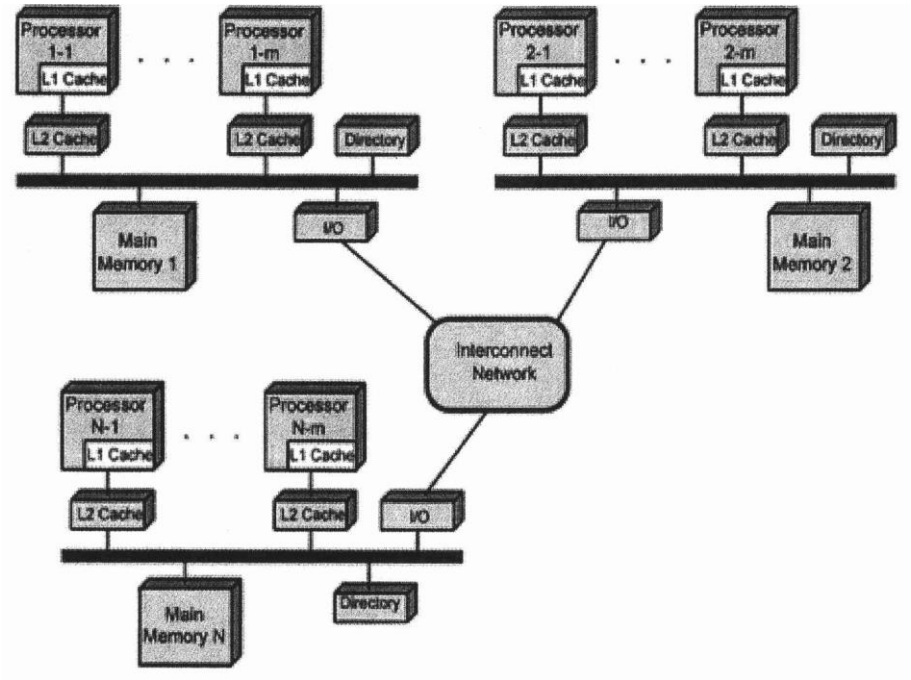



*Figure 1: NUMA Architecture*

**QUESTION 2** (5 marks)

Question 1 is based on the following Figure 2.
Inventor 'Y' was France's most
celebrated
mathematician
and
physicist. In 1642, inventor 'Y'
invented
and
build
the
first
working calculating machine as a
means of helping his father to

perform tedious tax accounting.

a) Who is inventor 'Y'?

b) Name another calculating machine that was invented in the same generation as inventor Y's calculating machine.

c) Which programming language is named after inventor 'Y'?

d) State TWO (2) mathematical operationsthat can be performed byinventor Y's calculating machine.

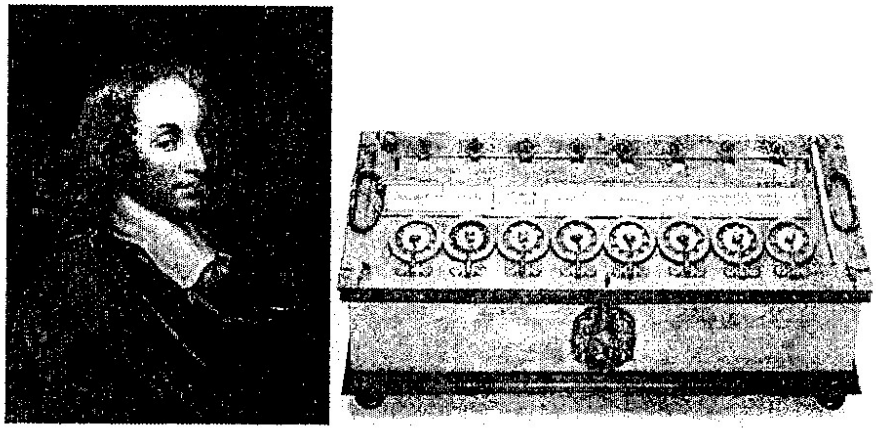


*Figure 2: Inventor 'Y' and his calculating machine.*

**QUESTION 3** (5 marks)

Calculate -9Brr - 37a using 1's and 2's complement method. Show your complete workflow.

**QUESTION 4** (5 marks)

a) What is mapping function?

b) Differentiate between direct and associative mapping in cache.

c) If a 28-bit address is used with word size = 4 bit and line = 10 bit, calculate the tag size of both direct and associative mapping.

**QUESTION 5** (10 marks)

a) Differentiate between interrupt-driven 1/0 and programmed 1/0.

b) Discuss ONE (1) benefit of programmed 1/0.

c) Discuss TWO (2) disadvantages of multiple interrupts.

**QUESTION 6** (10 marks)

Question 10 is based on Figure 3. Figure 3 depicts a Non-uniform Memory Access (NUMA) architecture.

a) Redraw a symmetric multiprocessor organization based on Figure 3.

b) Give TWO (2) examples of tightly coupled Multiple Instruction, Multiple Data (MIMD)

parallel processor architectures.

c) Discuss TWO (2) disadvantages of tightly coupled Multiple Instruction, Multiple Data (MIMD) parallel processor architectures.

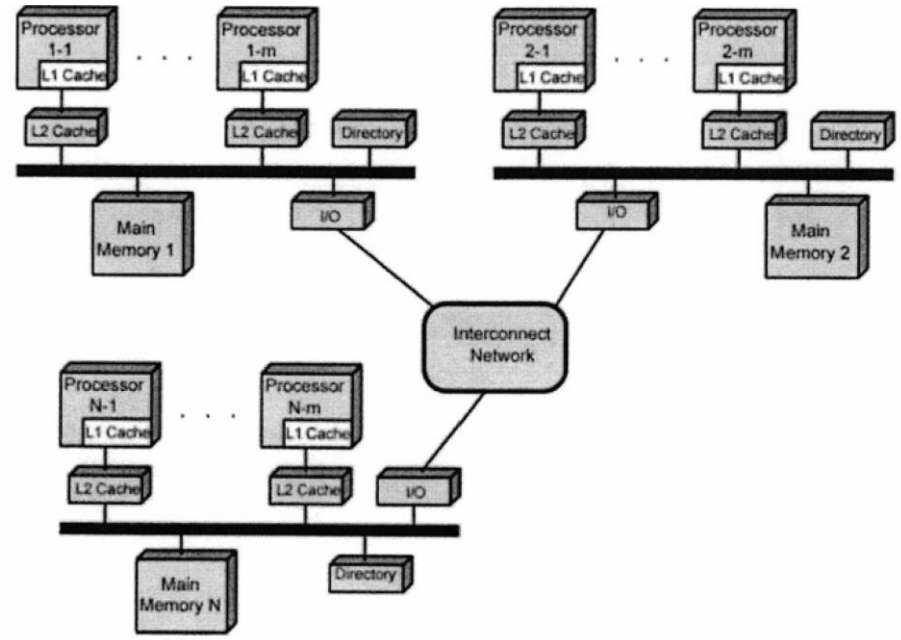



*Figure 3: Non-uniform 'Memory Access (NUMA)*

**QUESTION 7** (10 marks)

a) Draw Single Instruction, Single data (SISD) architecture with the following information:
- CU is the control unit
- PU is the processing unit
- MU is the memory unit
- IS is the instruction stream
- DS is the data stream

b) Describe the parallel architecture.

c) Explain ONE (1) benefit of parallel architecture.

d) Differentiate between loosely-coupled and strongly-coupled Multiple Instruction, Multiple Data stream (MIMD) architecture with example.

**QUESTION 8** (10 marks)

a) Given x = AA + CD; show the workflow to solve the equation in:

i. Hexadecimal value
Decimal value

b) Given y = IA - 3D; show the workflow to solve the equation in:

i. Sign and magnitude representation
I's and 2's complement representation

**C** Hak Cipta Universiti Teknologi MARA

**QUESTION 9** (10 marks)

a) Discuss THREE (3) types of interfaces to external devices.

b) Classify the following input/output devices to their respective 1/0 module:

i. Network interface card
Scanner
Touch Screen

i. Register

**QUESTION 10** (10 marks)

a) Discuss the TWO (2) processes in the machine cycle that are governed by the main memory.

b) List TWO (2) operations that are comprised in the Execute phase.

c) Explain what happens in the Program Counter (PC) and Instruction Register (IR) during the Fetch cycle.

**PART B (20 MARKS)**

Answer the question in this part.

**QUESTION 11** (20 marks)

Question 1 is based on Figure 4 and Table 1.
Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 4 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Note that Register A, B, C, and D cannot be modified, and Register x, y, and z can be used as temporary registers, However, the final answer must be stored in Register y.

$$y = \frac{A^2 - (B + AC)}{AD}$$

*Figure 4*

*Table 1: Machine Instruction*

| Zero-address | One-address | Three-address |
|---|---|---|
| PUSH | LOAD | ADD |
| pop | STOR | SUB |
| ADD | ADD | MUL |
| SUB | SUB | DIV |
| MUL | | |
| DIV | DIV | |